Blueprint

Suggested overall length	Recommended range	Average length in my field
Benchmark	6,000 – 8,000 words (without references)	6,900 words (empirical ML/EWS papers)

Section	Element	Purpose	In my field?	My notes and ideas
Introduction *500 – 1000 words* **Introduction** *500 – 1000 words* **Introduction** *500 – 1000 words* **Introduction** *500 – 1000 words* **Introduction** *500 – 1000 words* **Introduction** *500 – 1000 words*	**Importance of the topic**	Introduce the topic and outline why it should be studied	☑ Yes ☐ No	Monsoon rainfall accounts for 75-85% of Nepal's rain; EDW and cloudbursts amplify deadly floods & economic losses.
	Brief literature review	Situate your study within the context of your field	☑ Yes ☐ No	Traditional EWS relies on downstream gauges with only 2-4 hrs lead time; modern satellite ML (GPM/CHIRPS/ERA5) and AI streamflow models (Nearing et al., Nature 2024) enable predictive forecasting.
	Research gap	Provide justification for your study	☑ Yes ☐ No	Sparse gauge density, complex Himalayan orography, and lack of coupled deep learning rainfall-soil saturation forecasting in Nepal (Bhattarai et al., 2024).
	Research aim	Present what your study was about	☑ Yes ☐ No	Build spatiotemporal deep learning (NowcastNet-inspired / TFT) models to extend proactive flood lead time to 12-24 hrs.
	The main contributions of the paper (optional)	Highlight what novel insights your study offers	☑ Yes ☐ No	1) 23-yr merged dataset, 2) First multi-horizon deep learning applied to Nepal transboundary basins, 3) Dynamic RPI alert matrix coupling rainfall & soil moisture.
	Structure of the paper (optional)	Present the organization of the paper	☑ Yes ☐ No	Organized into 5 main sections: related work, dataset/methodology, experimental results, early warning evaluation, and conclusion.
Literature Review *optional · 1000 – 1500 words* **Literature Review** *optional · 1000 – 1500 words* **Literature Review** *optional · 1000 – 1500 words*	**Definitions of key concepts**	Provide any necessary definitions of contested terms important in your research	☑ Yes ☐ No	Cloudburst (>50 mm/hr or >100 mm/24h), Antecedent Precipitation Index (API), Runoff Potential Index (RPI), Elevation-Dependent Warming.
	Key themes from the literature	Outline the literature on the main topics related to your research	☑ Yes ☐ No	Transboundary flood susceptibility in Nepal (Bhattarai et al., 2024); AI

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				global ungauged flood prediction (Nearing et al., 2024); Physics-ML extreme rainfall nowcasting (Zhang et al., Nature 2023).
	Critical analysis	Analyze previous studies critically to reinforce the research gap	☑ Yes ☐ No	Traditional hydrological models (GloFAS/WRF) require dense calibration and struggle in ungauged Himalayan terrain; recent AI models outperform them but require local orographic adaptation.
Theoretical Framework *optional · 200 – 500 words*	**The theories utilized in the study**	Present which theories you drew upon when designing the study and/or analyzing the data	☑ Yes ☐ No	Orographic precipitation lifting dynamics, Clausius-Clapeyron scaling (7%/°C moisture capacity), hydrologic routing concentration time.
Methodology *1000 – 1500 words* **Methodology** *1000 – 1500 words* **Methodology** *1000 – 1500 words* **Methodology** *1000 – 1500 words* **Methodology** *1000 – 1500 words*	**Definition and justification of the research methodology**	Outline and justify the chosen methodology	☑ Yes ☐ No	Spatiotemporal deep generative nowcasting (NowcastNet) for 2D cloud tracking coupled with catchment TFT/LSTM for multi-horizon accumulated precipitation and runoff.
	Background information about context (optional)	Present the context in which your study was conducted (e.g. geography)	☑ Yes ☐ No	All of Nepal across 5 major basins (Koshi, Gandaki, Narayani, Karnali, West Rapti), elevations from 60m to 8848m.
	Sample & data sources + ethical considerations	Outline who/what you studied, how you got it, and any ethical considerations	☑ Yes ☐ No	2001-2023 DHM ground stations, NASA GPM IMERG 0.1°, CHIRPS 0.05°, ERA5-Land 0.1° reanalysis, HydroSHEDS basin shapefiles. Fully open/cited.
	Research tools and procedures	Specify which research tools you used and specifically how you conducted study	☑ Yes ☐ No	Python (PyTorch, PyTorch-Forecasting, GeoPandas, Xarray), Google Earth Engine (GEE) preprocessing, quantile mapping station bias correction.
	Data analysis techniques	Present how you analyzed your data and justify your choice	☑ Yes ☐ No	Rolling-origin temporal cross-validation (train: 2001-2016, val: 2017-2019, test: 2020-2023). Moving-window lag features for baselines.
Results *1500 – 3000 words* **Results** *1500 – 3000 words*	**Presentation of the main results**	Present the results you obtained	☑ Yes ☐ No	Deep learning models achieve superior accuracy (RMSE 4.8 mm, KGE 0.84) across 6h, 12h, 24h, 48h horizons, significantly exceeding classical baselines across all basins.
	Clear use of tables, figures or quotes	Provide a visual representation of the data and raw evidence	☑ Yes	Fig 1: Orographic elevation cross-section; Fig 2:

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			☐ No	Architecture diagram; Fig 3: Baseline performance table; Fig 4: Dynamic 2D alert matrix & ROC.
Discussion *500 – 1000 words* **Discussion** *500 – 1000 words* **Discussion** *500 – 1000 words* **Discussion** *500 – 1000 words*	**Restatement of result to be discussed**	Remind the reader of the result	☑ Yes ☐ No	Spatiotemporal models capture convective storm trajectories while attention mechanisms identify 3-5 day antecedent soil moisture memory.
	Comparison of result with literature	Situate the result within the literature context	☑ Yes ☐ No	Corroborates Nearing et al. (Nature 2024) on ungauged basins and Bhattarai et al. (2024) on Gandak flood susceptibility, proving DL resilience under extreme Himalayan terrain.
	Explanation of findings and/or differences	Explain findings obtained and possible differences with previous studies	☑ Yes ☐ No	Satellite-only inputs underestimate localized cloudbursts without ground station quantile mapping bias correction.
	Interpretation of the findings	Show what your findings mean and how they can be understood	☑ Yes ☐ No	Coupling upstream forecast with antecedent soil saturation shifts early warning from reactive river-stage alarm to proactive evacuation.
Conclusion *500 – 1000 words* **Conclusion** *500 – 1000 words* **Conclusion** *500 – 1000 words* **Conclusion** *500 – 1000 words* **Conclusion** *500 – 1000 words*	**Restatement of main aim and topic**	Remind the reader of the aim and topic of your study	☑ Yes ☐ No	Investigated ML time-series forecasting of monsoon rainfall patterns in Nepal to optimize flood early warning lead times.
	Recap of key findings & contributions	Restate main results and discuss briefly	☑ Yes ☐ No	Extended actionable evacuation lead times by 12-24 hours with CSI > 0.75, FAR < 0.20, and validated against historical 2017 & 2021 disasters.
	Practical implications of findings	Show what impact findings have for different stakeholders	☑ Yes ☐ No	Directly operationalizable by DHM and NDRRMA for targeted municipal alerts, mitigating agricultural and infrastructure damage.
	Limitations	Acknowledge the limitations of your study	☑ Yes ☐ No	Sparse radar coverage in Nepal, high-elevation gauge freeze-out in Himalayas, latency in near-real-time satellite product availability.
	Suggestions for future research	Suggest what future research should focus on	☑ Yes ☐ No	Integrate radar nowcasting grids, physics-informed neural networks (PINN) for runoff routing, and mobile edge deployment for SMS warnings.